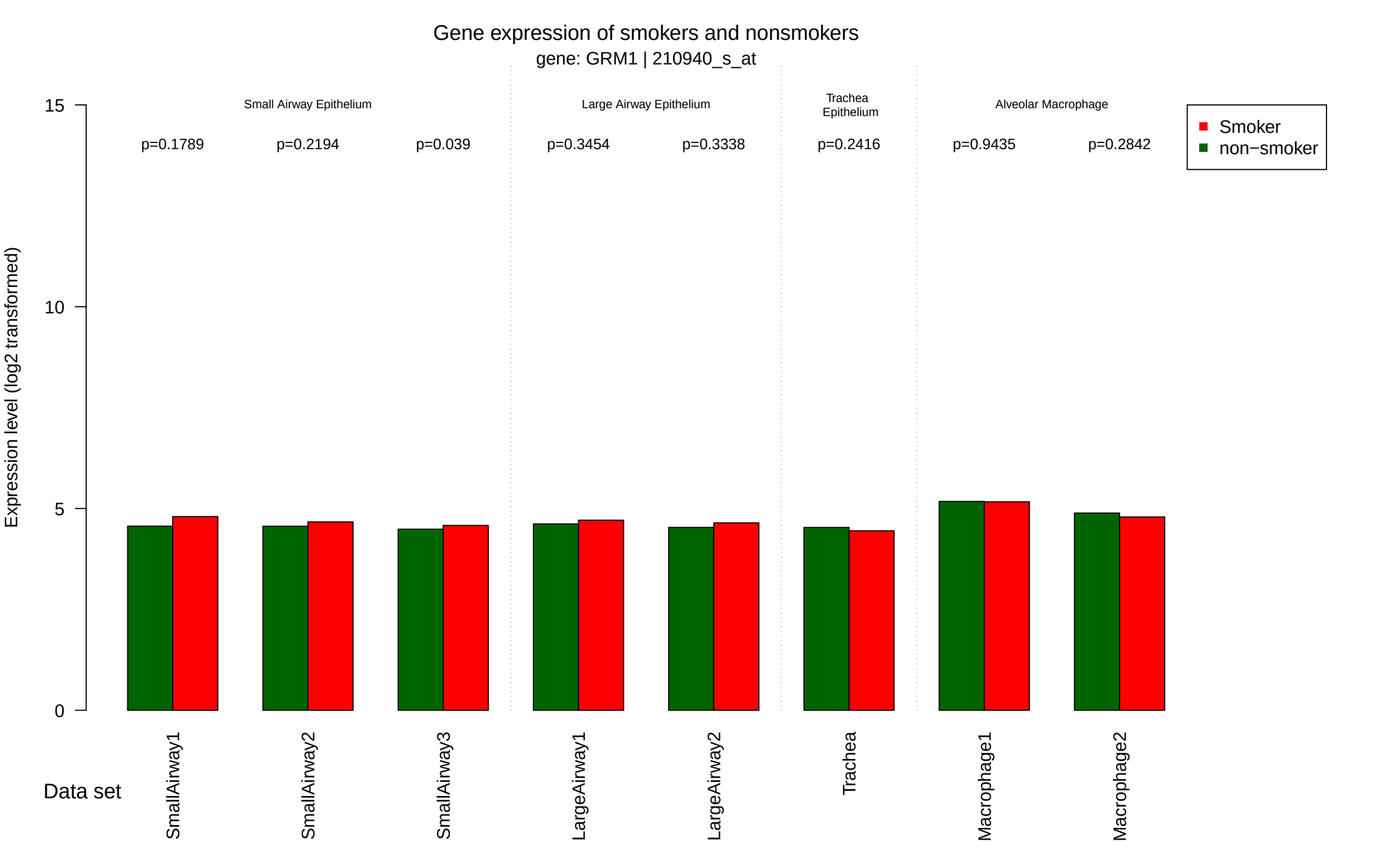
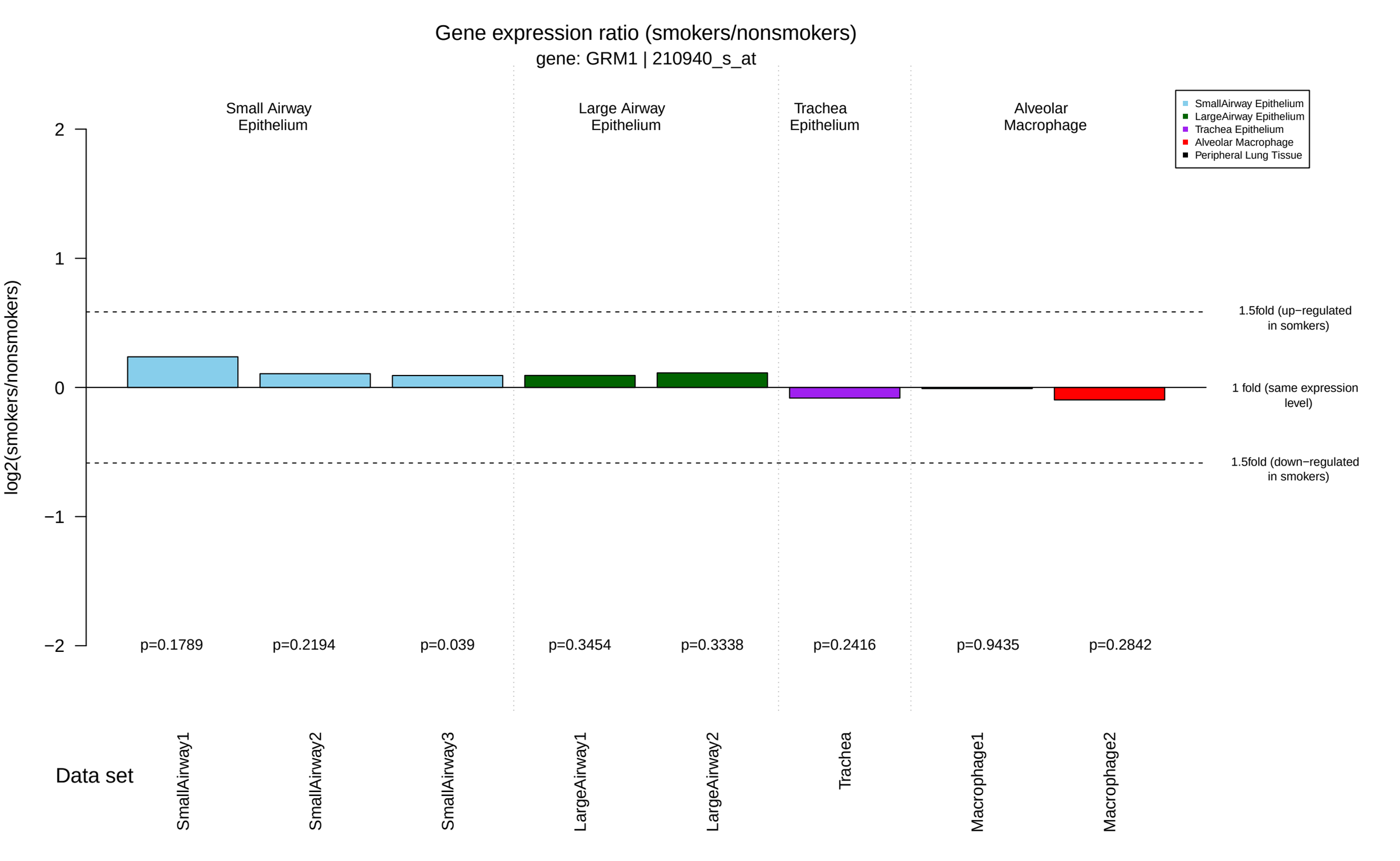
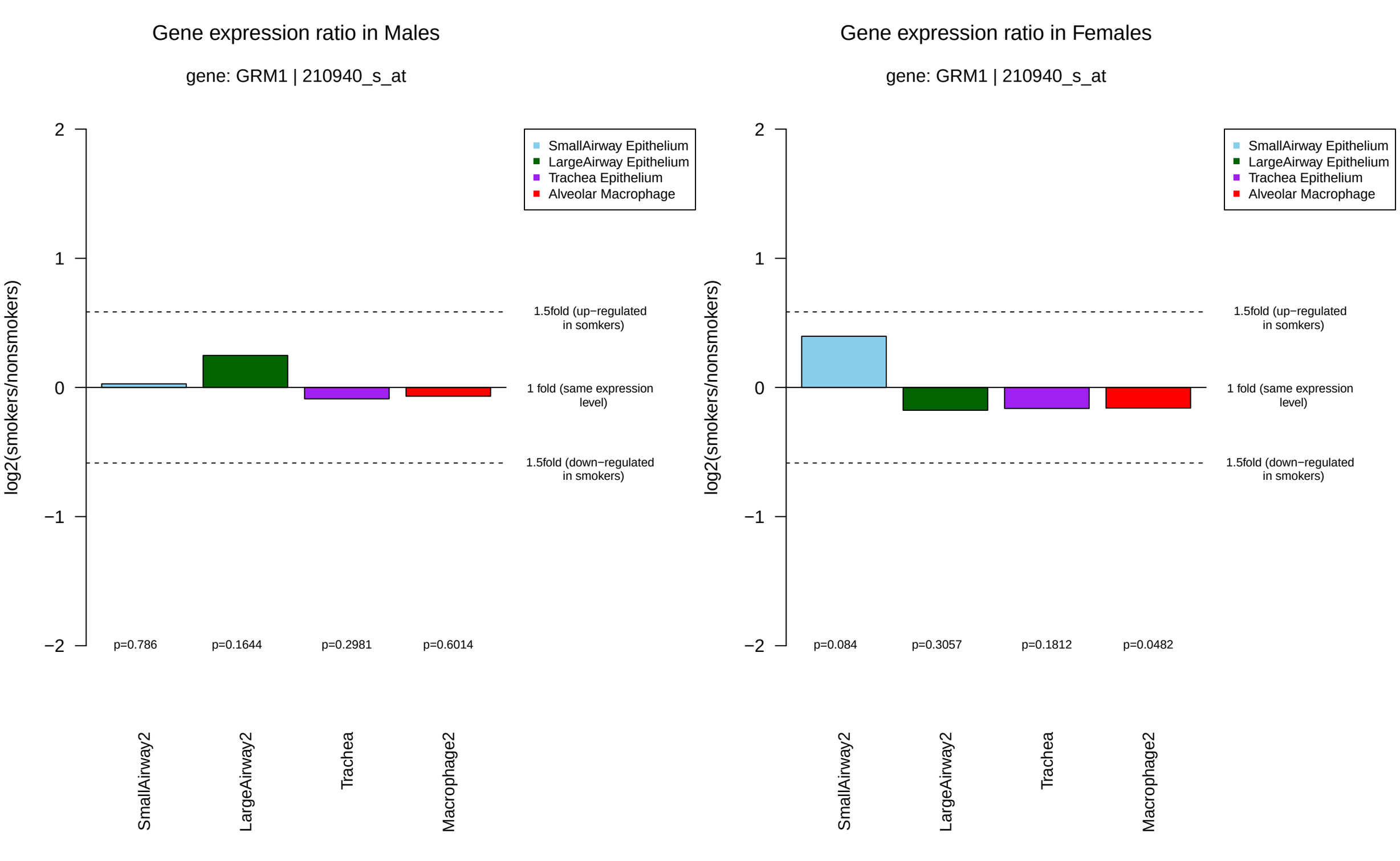
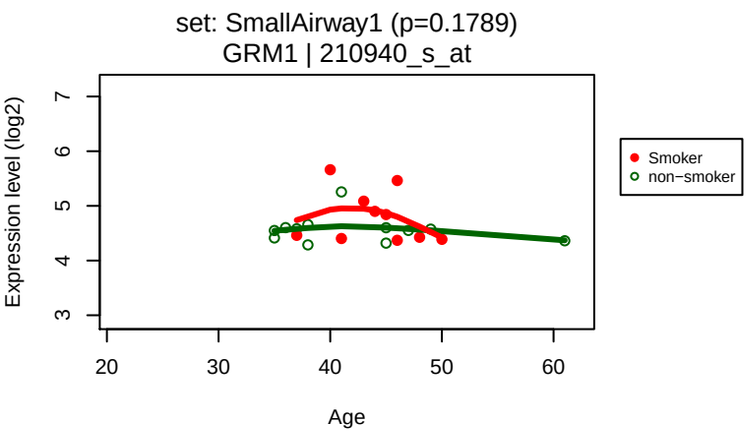
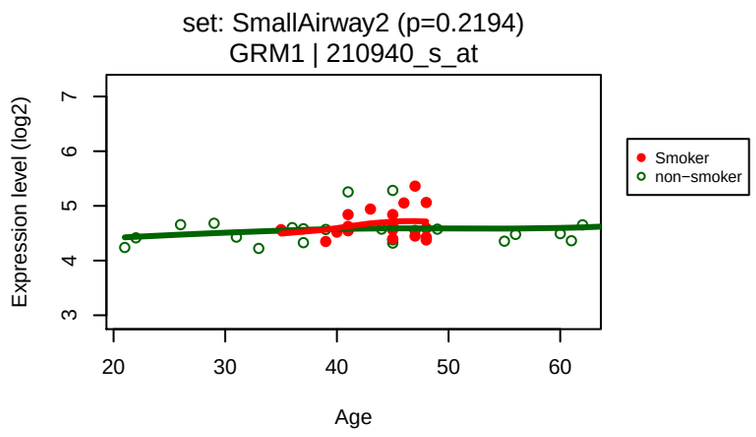
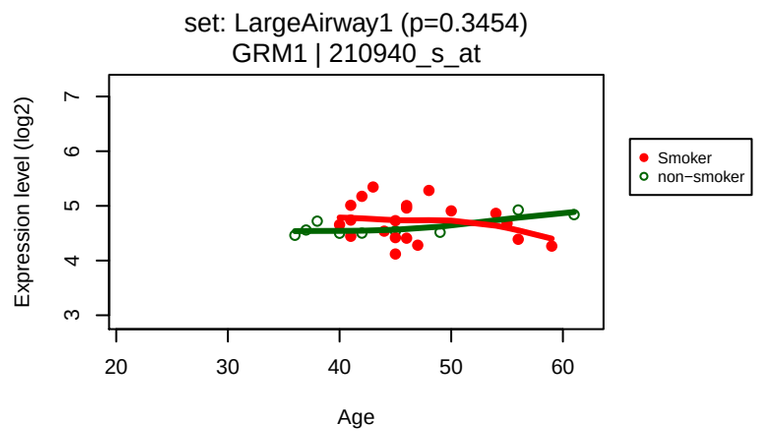

Gene expression of smokers and nonsmokers
gene: GRM1 | 210940_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1789
p=0.2194
p=0.039
p=0.3454
p=0.3338
p=0.2416
p=0.9435
p=0.2842
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

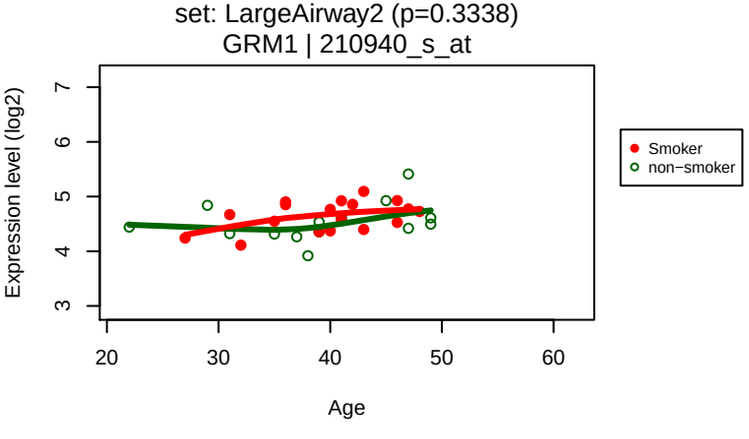

Gene expression ratio (smokers/nonsmokers)
gene: GRM1 | 210940_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1789
p=0.2194
p=0.039
p=0.3454
p=0.3338
p=0.2416
p=0.9435
p=0.2842
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

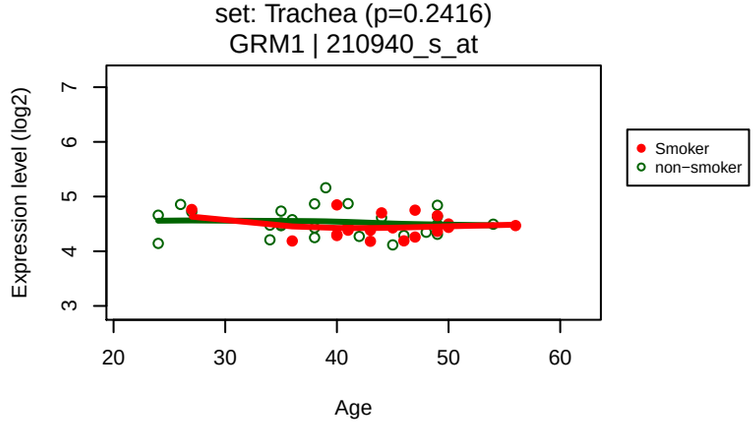

Gene expression ratio in Males
gene: GRM1 | 210940_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.786
p=0.1644
p=0.2981
p=0.6014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM1 | 210940_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.084
p=0.3057
p=0.1812
p=0.0482
SmallAirway2
LargeAirway2
Trachea
Macrophage2

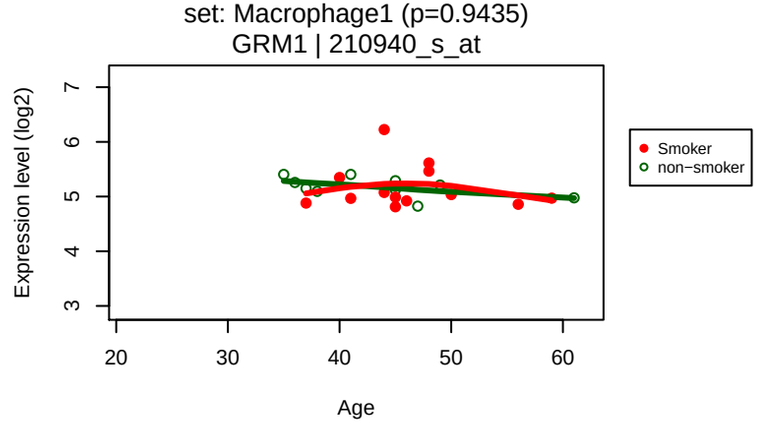

set: SmallAirway1 (p=0.1789)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker

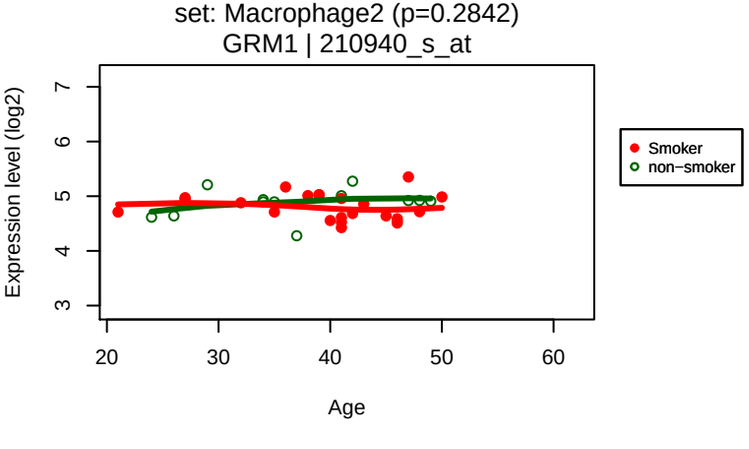

set: SmallAirway2 (p=0.2194)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker

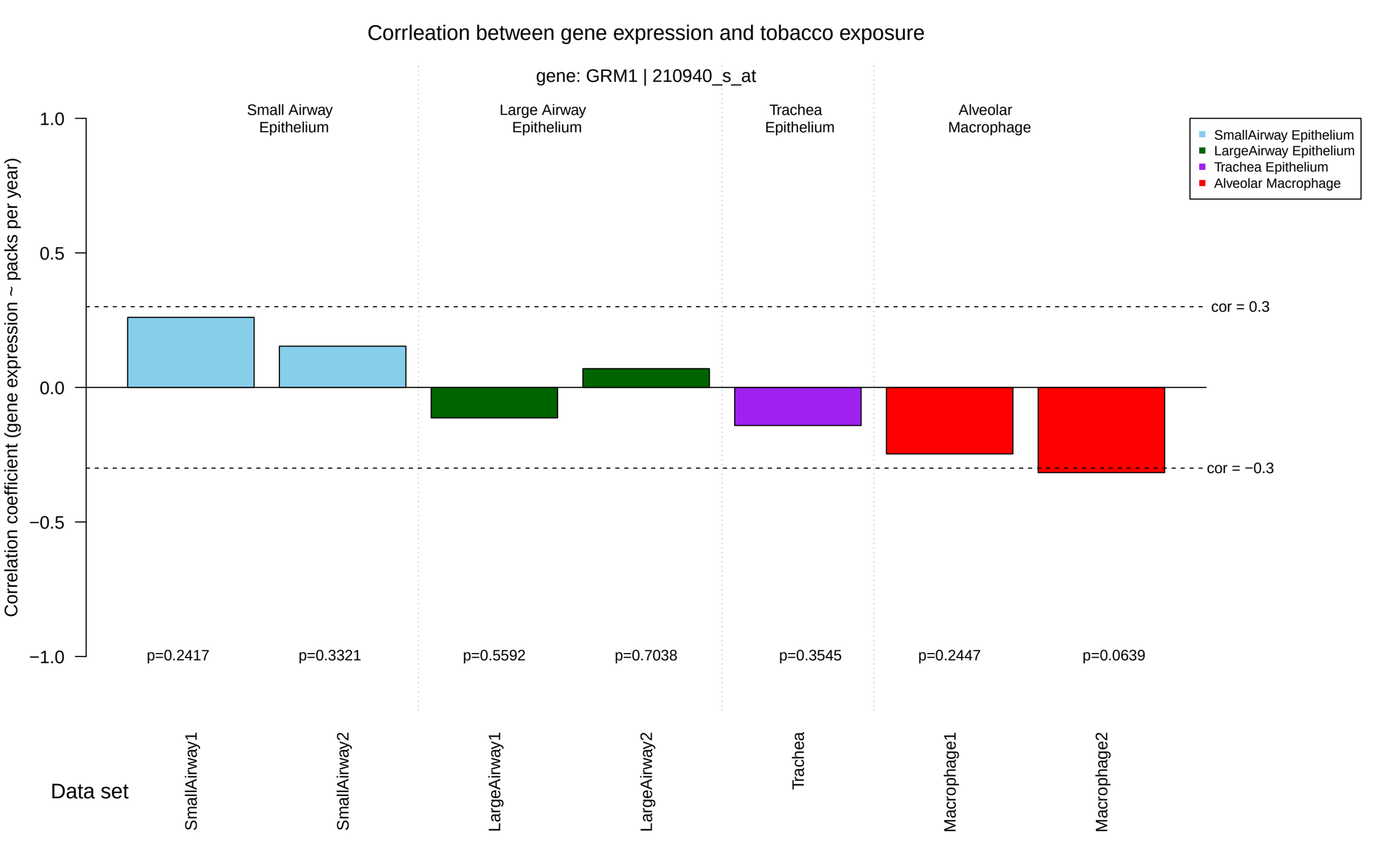

set: LargeAirway1 (p=0.3454)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3338)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2416)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9435)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2842)
GRM1 | 210940_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM1 | 210940_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2417
p=0.3321
p=0.5592
p=0.7038
p=0.3545
p=0.2447
p=0.0639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2